

SESSION 2

Classic Machine Learning Overview

Regression • Classification • Clustering • Dimensionality Reduction • Forecasting • Anomaly Detection

Applied AI Design Workshop for Faculty

Fred Livingston (fjliving@ncsu.edu) 2026

Goal: match a framed disciplinary problem to the right class of classical ML methods.

Session focus

- ML workflow
- Supervised vs. unsupervised
- Six core task families
- Scenario sorting activity
- Discussion on model fit

Workshop outcomes and Session 2 role

Session 2 gives faculty the “model-family map” needed after Session 1 problem framing and before regression/classification fundamentals.

**Translate
Disciplinary
question → ML task**

**Differentiate
Regression,
classification,
clustering,
forecasting**

**Baseline
Know what a
simple model
must beat**

**Evaluate
Link task type to
first metrics**

**Select
Choose method
families before
tools**

Design principle: the question determines the learning task; the learning task constrains the models and metrics.

Day 3 schedule and flow

From engineering problem to ML problem

Done

Classic ML overview

Now

Invited presentation: From Prompts to Agents

Context

Break

Regression fundamentals

Classification fundamentals

Methods

Hands-on baseline model activity

Practice

Learning activities for Session 2



Participant output: a task-family recommendation for each scenario plus one “why this task, not another?” explanation.

Classic ML workflow: from problem to evidence



Classic ML emphasizes structured data, explicit targets or patterns, measurable performance, and reproducible comparison to baselines.

Key question for today: where does your problem sit in this workflow?

Supervised and unsupervised learning

Supervised learning

Uses examples with known targets.

Typical tasks:

- **Regression**
- **Classification**
- **Forecasting**

Core question:

Can the model learn from labeled examples to predict a new case?

Unsupervised learning

Uses data without known targets.

Typical tasks:

- **Clustering**
- **Dimensionality reduction**
- **Pattern discovery**

Core question:

Can the model reveal structure that merits domain interpretation?

Boundary case: anomaly detection can be supervised, semi-supervised, or unsupervised depending on whether labeled abnormal cases exist.

Regression: predicting continuous outcomes

Use regression when the target is a number and error magnitude matters.

Questions

- How much?
- How many?
- What value?

Outputs

- Continuous prediction
- Prediction error

Common methods

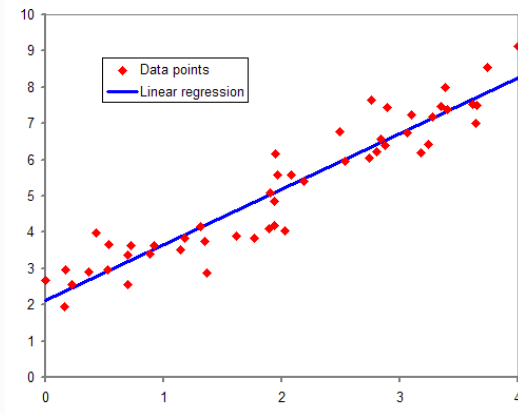
- Linear regression
- Polynomial regression
- Ridge / Lasso
- Regression trees
- Random forests

First metrics

- MAE
- RMSE
- R^2

Baseline

- Predict mean/median



Source: https://en.wikipedia.org/wiki/Regression_analysis#/media/File:Normdist_regression.png

Examples: final material strength, energy consumption, cycle time, bioprocess yield, exam score, remaining useful life.

Classification: predicting categories

Use classification when the output is a discrete category, label, or decision class.

Questions

- Which class?
- Pass or fail?
- Fault or normal?

Outputs

- Class label
- Probability

Common methods

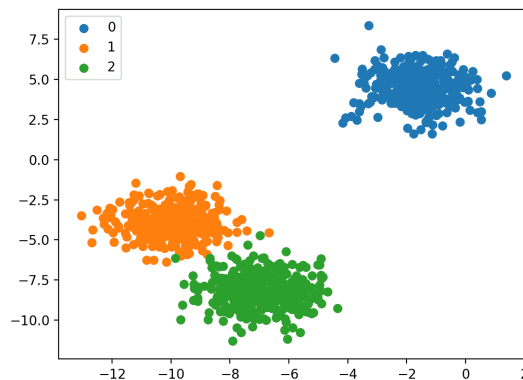
- Logistic regression
- Decision trees
- k-nearest neighbors
- Support vector machines
- Random forests

First metrics

- Accuracy
- Precision
- Recall
- F1-score

Baseline

- Majority class



Source: <https://machinelearningmastery.com/types-of-classification-in-machine-learning/>

Examples: product accept/defect, medical screening, equipment fault state, material type, student support category.

Clustering: finding groups without labels

Use clustering when the goal is to explore unlabeled data and identify possible groups.

Questions

- What groups exist?
- Are there operating regimes?
- Are there natural profiles?

Outputs

- Cluster ID
- Similarity grouping

Common methods

- k-means
- Hierarchical clustering
- DBSCAN

Preconditions

- Meaningful features
- Appropriate scaling

Evaluation

- Cluster separation
- Stability
- Domain interpretation

Caution

- Clusters are not automatically scientific categories

Examples: operating regimes, product families, student learning profiles, patient subgroups, sensor behavior patterns.

Dimensionality reduction: simplifying many variables

Use dimensionality reduction when many correlated variables need to be summarized or visualized.

Questions

- Can many variables be summarized?
- What structure is visible in 2D?
- Which directions explain variation?

Common methods

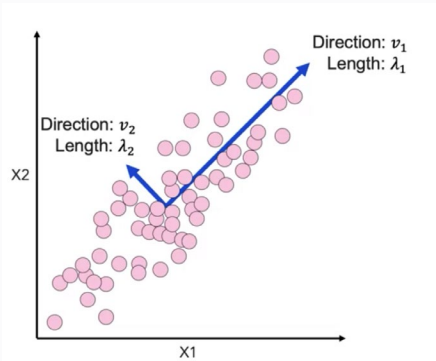
- Principal component analysis
- t-SNE / UMAP for visualization

Outputs

- Components
- Low-dimensional projections

Use carefully

- Components may be hard to interpret
- Scaling matters
- Visualization is not proof of separability



Source: <https://medium.com/the-quant-journey/unsupervised-machine-learning-dimensionality-reduction-and-pca-8ed5db205200>

Examples: compress sensor streams, visualize high-dimensional experiments, reduce multicollinearity, explore image-derived features.

Forecasting: predicting future values

Use forecasting when time order matters and future values depend on past behavior.

Questions

- What happens next?
- What will demand be tomorrow?
- How will a process variable evolve?

Common features

- Lags
- Rolling means
- Trends
- Seasonal indicators
- Time-window statistics

Validation

- Preserve time order
- Train on past, test on future
- Avoid random split leakage

Examples: energy demand, production throughput, sensor trends, inventory needs, process yield over time.

Anomaly detection: identifying unusual observations

Use anomaly detection when rare, unexpected, or abnormal behavior is the focus.

Questions

- Is this unusual?
- Does this run deviate from normal?
- Which cases need review?

Common methods

- Thresholds
- Isolation forest
- One-class methods
- Distance-based methods
- Control-chart-inspired rules

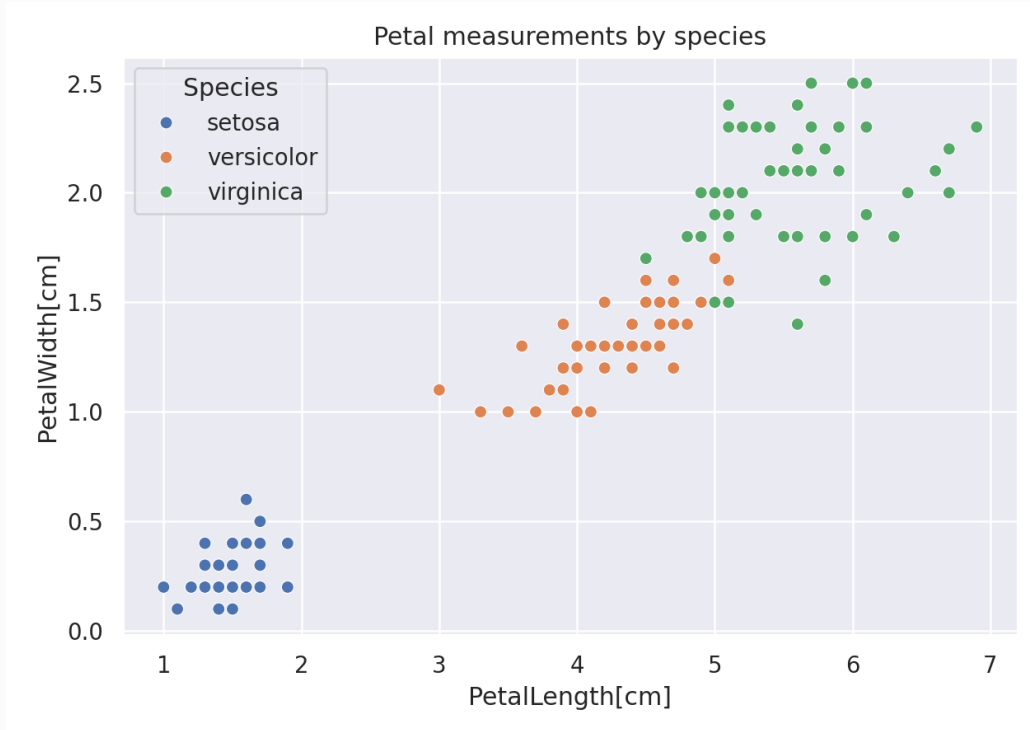
Practical issues

- Rare labels
- False alarms
- Changing normal behavior
- Need for human review

Examples: equipment condition monitoring, lab-sensor drift, cybersecurity events, defective runs, unusual student activity patterns.

Session 2 technical skill: visualize before modeling

Goal: use quick plots to understand relationships, distributions, and target balance.



```
import seaborn as sns
import matplotlib.pyplot as plt

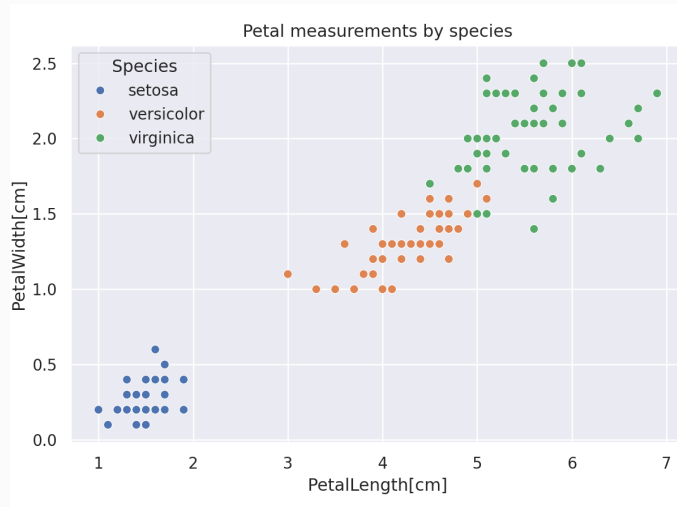
sns.scatterplot(data=df,
                x="PetalLength_cm",
                y="PetalWidth_cm",
                hue="Species")
plt.show()
```

- Do classes appear separable?
- Are there obvious outliers?
- Is the relationship linear or nonlinear?
- What should not be concluded from the plot?

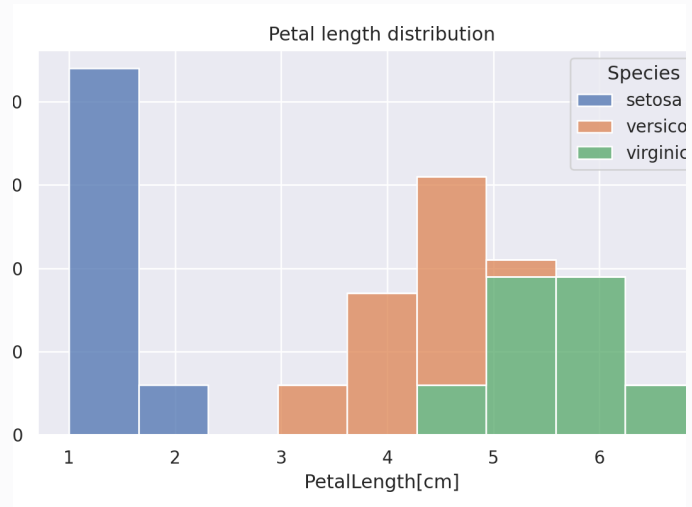
[Session 2 notebook](#)[02_session2_visualization.ipynb](#)

Three reuse throughout the workshop

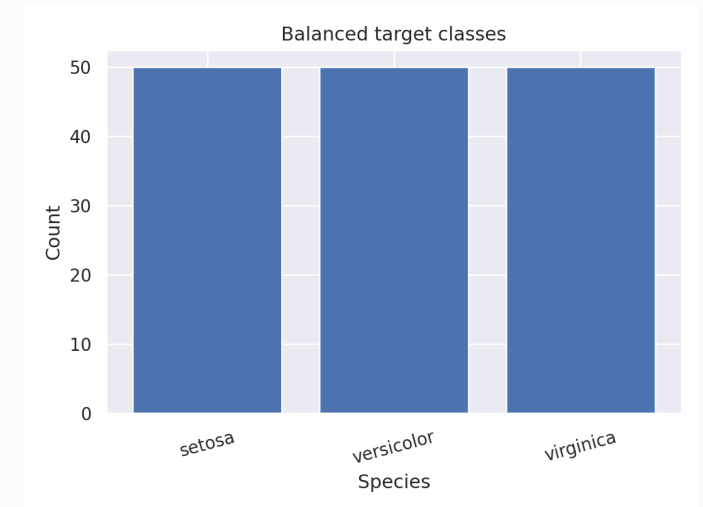
Use only the plots that support the applied modeling decision.



Scatterplot
relationship + separability



Histogram
distribution + overlap



Count plot
target balance

Instructor prompt

“Use the plot to decide what question to ask next. Do not treat visualization as proof of causality or final model performance.”

Task chooser: match the question to the task

Task	Question	Data signal	Early metric or check
Regression	How much?	Continuous target	MAE / RMSE / R^2
Classification	Which category?	Known labels	Precision / recall / F1
Clustering	What groups?	No target label	Separation + interpretation
Dimensionality reduction	Can we simplify?	Many variables	Explained variance
Forecasting	What next?	Time-ordered data	Future prediction error
Anomaly detection	What is unusual?	Rare or unknown events	False alarms / missed events

Activity: scenario sorting

For each scenario, identify the ML task and defend the choice in one sentence.

A. Predict final strength from process parameters.

B. Determine whether an image shows abnormality.

C. Identify natural groups of participants.

D. Summarize 50 sensor variables into fewer components.

E. Predict next week's energy demand.

F. Flag unusual laboratory measurements.

Debrief: Which scenarios were ambiguous? What extra information would resolve the task choice?

Classroom discussion prompts

Can the same dataset support more than one task?

What task requires labeled data?

When is clustering scientifically meaningful?

How is forecasting different from ordinary regression?

How is anomaly detection different from fault classification?

What does classic ML provide that an LLM alone does not?

Discussion goal: make model choice a transparent disciplinary argument, not a default software selection.

Session 2 outputs and transition

You should leave with:

- 1 A task-family choice for their framed problem
- 2 A reason the choice matches the research or teaching question
- 3 A first baseline idea
- 4 One evaluation metric or qualitative check
- 5 One ambiguity or missing data item to resolve

Next: Regression fundamentals — predicting continuous disciplinary outcomes and interpreting error.